📄 Project 2: Wazuh SIEM Implementation

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📅 Duration: 4 Weeks

📌 Week 1: Installation & Setup

🎯 Objective:

To install and configure the **Wazuh SIEM platform** on an Ubuntu system.

🛠️ Tasks Performed:

- 🔹 **Installed Required Dependencies:**
  - **Docker**
  - **curl**
  - System packages

- 🔹 **Fixed System Issues:**
  - Installed missing curl
  - Updated system using:

```
sudo apt install curl -y
sudo apt update && sudo apt upgrade -y
```

🔹 **Installed Wazuh:**

- Downloaded installation script
- Resolved OS compatibility issue using:

```
sudo bash wazuh-install.sh -a -i
```

🔹 **Components Installed:**

- Wazuh Manager
- Wazuh Dashboard
- OpenSearch

🔹 **Verified Installation:**

```
sudo systemctl status wazuh-manager
```

Opened dashboard:

```
https://localhost
```

📊 **Architecture Diagram**

```
User System
     ↓
Wazuh Agent
     ↓
Wazuh Manager
     ↓
Wazuh Dashboard
     ↓
Security Alerts & Logs
```

⚠️ Challenges Faced

- OS not supported
- Missing curl
- Installation errors

✅ Solutions

- Installed dependencies
- Updated system
- Used ignore flag

🎯 Outcome

✔️ Wazuh installed successfully
✔️ Dashboard accessible
✔️ System ready for monitoring

📌 Week 2: Configuration & Monitoring

🎯 Objective:

To configure Wazuh for real-time monitoring and generate alerts.

🛠️ Tasks Performed:

🔹 **Configuration:**

- Edited config file:

```
sudo nano /var/ossec/etc/ossec.conf
```

🔹 **Enabled Monitoring:**

- File Integrity Monitoring (FIM)
- Real-time monitoring
- Directory tracking:

```
/etc
/home
```

🔹 **Restarted Services:**

```
sudo systemctl restart wazuh-manager
```

🔹 **Testing:**

Created test file:

```
touch /home/mariyam/testfile.txt
echo "test" >>
/home/mariyam/testfile.txt
```

🔹 **Verified Alerts:**

- Opened dashboard
- Checked **Security Events**

- Observed file change alerts

🔍 Detailed Explanation

Week 2 focuses on **monitoring and detection**.

The **Wazuh SIEM system** works as:

- Agent collects logs
- Manager analyzes data
- Dashboard displays alerts

It detects:

- File changes
- System activity
- Suspicious behavior

📊 Monitoring Flow Diagram

```
File Change / Activity
          ↓
     Wazuh Agent
          ↓
     Wazuh Manager
          ↓
   Rule Processing
          ↓
   Alert Generated
          ↓
 Wazuh Dashboard
```

🎯 Outcome (Week 1 + Week 2)

✔️ SIEM system installed
✔️ Monitoring enabled
✔️ Real-time alerts working
✔️ Security visibility achieved

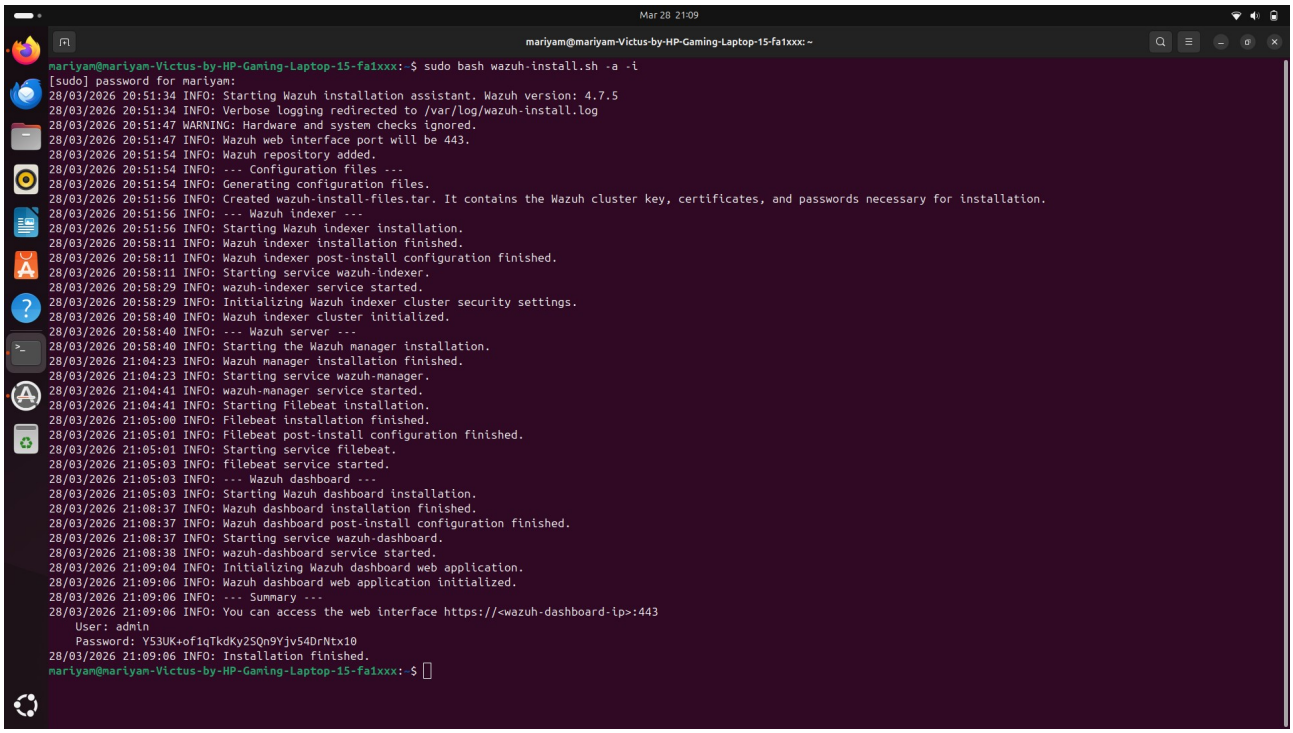
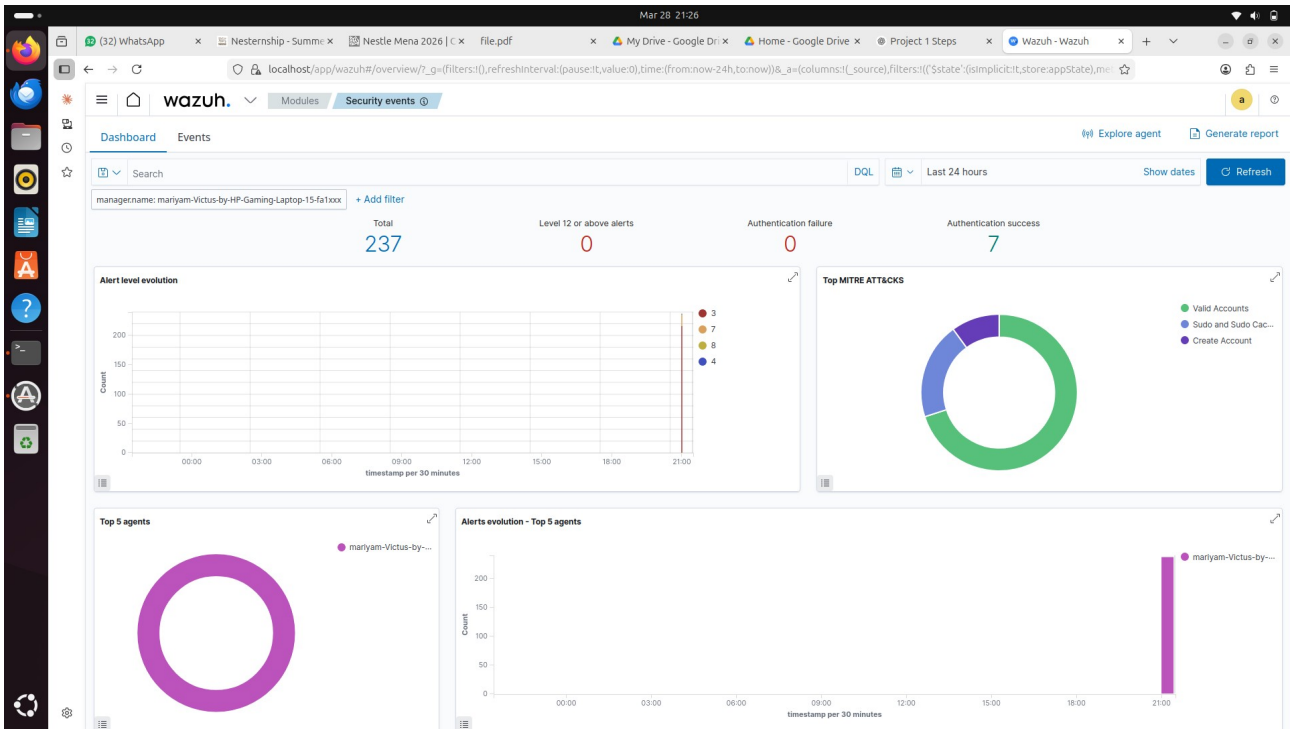
📌 Final Summary

This project demonstrates how **Wazuh** can be used to:

- Monitor system activity
- Detect threats
- Improve security

It builds a strong foundation for **SOC (Security Operations Center)**.

Wazu installation



Wazu Dashbord